Final Report

Independent Stave Company:
Sustainability Report (Oak Barrels)

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Table of Contents

1. Comparison of Vision and Risk Narrative 3

2. SDG Impact Value Map 5

3. SDG Impact Analysis.....7

4. System Redesign Value Map.....8

5. Comparison of Business Sustainability Strategy Narrative.....10

6. Business and Product/Service Innovation Narrative.....12

7. Types of Goods and Market Failures Narrative 15

8. What is a business for.....17

9. Assessment and Reporting 19

10. Stakeholders and Rights Holders Narrative.....21

11. Organizational Change.....24

12. Recommendations.....26

References.....27

Appendix A: PESTLE Analysis.....32

Appendix B: Quillbot AI Detection Results (By Sections) 38

Appendix C: Grammarly AI Detection Results (Full Report).....63

1. Comparison of Vision and Risk Narrative

Independent Stave Company's Sustainability Vision:

Independent Stave Company (ISC) is a family owned business committed to the long-term sustainability of oak forests. To be competitive in their line of work, ISC must adhere to the standards set by the U.S. Code of Federal Regulations, requiring all barrels classified as bourbon barrels to be made from new oak staves. Due to this requirement, ISC consumes oak trees at a considerable rate, but has committed to reforestation practices to ensure the oaks are replanted and available for generations to come.

ISC's sustainability approach extends beyond reforestation practices. For each oak purchased, the company uses 100% of the tree product. (Independent Stave Company, n.d.).Wood that does not go into stave, heading, or oak alternative material is utilized internally as fuel or sold as high quality by-products to be used in other industries.

The company has made declarations about becoming a more sustainable cooperage company and decreasing their carbon footprint. In an effort to do so, ISC has partnered with reputable sustainability organizations like American Forests, White Oak Initiative, Missouri Forest Products Initiative, and American Tree Farm System. Each of these organizations is shaped by standards and practices geared towards ethical and sustainable companies. Internally, ISC has created a Sustainability Division dedicated to advancing the way ISCO manages forests, materials, emissions, our workforce, and communities.

Our Vision:

- Barrel reuse and reconditioning (barrels as a service)
- Replacement of steel hoops with a more sustainable material
- Forest resource management
- Full system clean energy
- Community engagement

Independent Stave Company envisions a cooperage industry built on circular value rather than linear consumption, where barrels are reused, reconditioned, and managed as long-term assets

throughout their lifecycle. By coupling barrel-as-a-service models with responsible forest resource management, ISC seeks to reduce raw material pressure while ensuring the long-term health and regeneration of oak forests. By committing to using recycled steel and greenkote on the steel hoops, ISC will dramatically cut its carbon emission per barrel. The company aspires to operate within a fully integrated clean-energy system that lowers emissions across production, reconditioning, and distribution. With sustained investment in workforce development and local partnerships, ISC aims to strengthen the communities that support its forests, mills, and cooperages while remaining economically resilient in a changing global market.

Risks:

As a cooperage, ISC faces inherent risks to their practice. Considering the requirement of raw materials, ISC is susceptible to oak supply constraints, tightening of forestry regulations, natural disasters, environmental degradation, and climate change (refer to Appendix A for a detailed PESTLE analysis). When examining their internal operations, they run the risk of skilled labor shortage. According to the USDA Forest Service, "The logging industry has been particularly affected; the median age of workers in the industry was approaching 50 years in 2017 and a workforce contraction of 13 percent is projected over the next decade." (Espinoza, 2020)

Beyond these foundational risks, the barrel industry is susceptible to market trends, including changes in consumer behavior, trade and tariff considerations, currency volatility, and general economic growth and recession. Finally, ISC's entire business and sustainability practice hinges on the requirements set forward by the U.S. Code of Federal Regulations. If those requirements were to shift, the entire model would need to be reimagined.

2. SDG Impact Value Map

<u>Deforestation</u>:

- The United States harvests roughly 36 million cubic meters of various types of hardwood per year. However, in the eastern regions, the regrowth exceeds the harvest rate, with growth-to-removal rations reported at 1.8:1 (USDA Forest Service FIA, 2023).
- A typical 53-gallon oak barrel requires between 0.6 and 0..8 cubic meters of oak timber. (USDA Forest Service, FIA, 2023).
- Logging and wood product manufacturing represent between 5–10% of total forest carbon stock removals annually in the U.S. These emissions, however, are partially offset over time through regrowth. (US EPA, Inventory of U.S. Greenhouse Gas Emissions and Sinks).

<u>Labor Rights</u>

- Logging is categorized as one of the most hazardous occupations in the U.S., with a fatality rate ranging from 82 to 97 deaths per 100,000 workers annually (U.S. Bureau of Labor Statistics (BLS), Census of Fatal Occupational Injuries (2022)).
- Workforce composition also presents vulnerabilities, as migrant and contract labor are common in forestry sectors, with up to 30% of forestry workers being foreign-born.

These arrangements can increase risks related to wage and compliance vulnerability. (ILO forestry sector reports; U.S. Bureau of Labor Statistics).

Air Pollution

- Forestry equipment, including diesel powered harvesters, log loaders, skidders and chainsaws produce CO_2, CO, NOx, VOCs, and PAHs. (EPA,1995).
- If oak staves are fully dried in a kiln, rather than partially air dried, energy demand increases substantially. Industrial kiln drying emits up to 340 kg of CO_2 per drying cycle and can be impacted by fuel source and moisture content. (EPA, 2023; U.S. Energy Information Administration [EIA], 2023). If air drying is not utilized, emissions can climb significantly.
- Though the process of barrel charring only lasts between 15-60 seconds, the combustion of 1kg of dry wood releases approximately 1.65 kg of CO_2. (EPA, 2023).

Transportation:

- The barrel supply chain typically involves multiple transportation stages and the use of freight trucks. These heavy-duty trucks emit on average 161.8 g CO_2 per ton-mile. (EPA, 2023).

Steel Hoops

- Steel production is one of the most carbon intensive processes in the realm of barrel materials. The global emissions intensity of crude steel production is approximately 1.85 kg of CO_2 per kg of crude steel. (World Steel Association, 2022).
    - If each barrel contains 8-11 kg of steel, associated emissions would range between 15-20 kg CO_2 per barrel (World Steel Association, 2022; IEA, 2023).

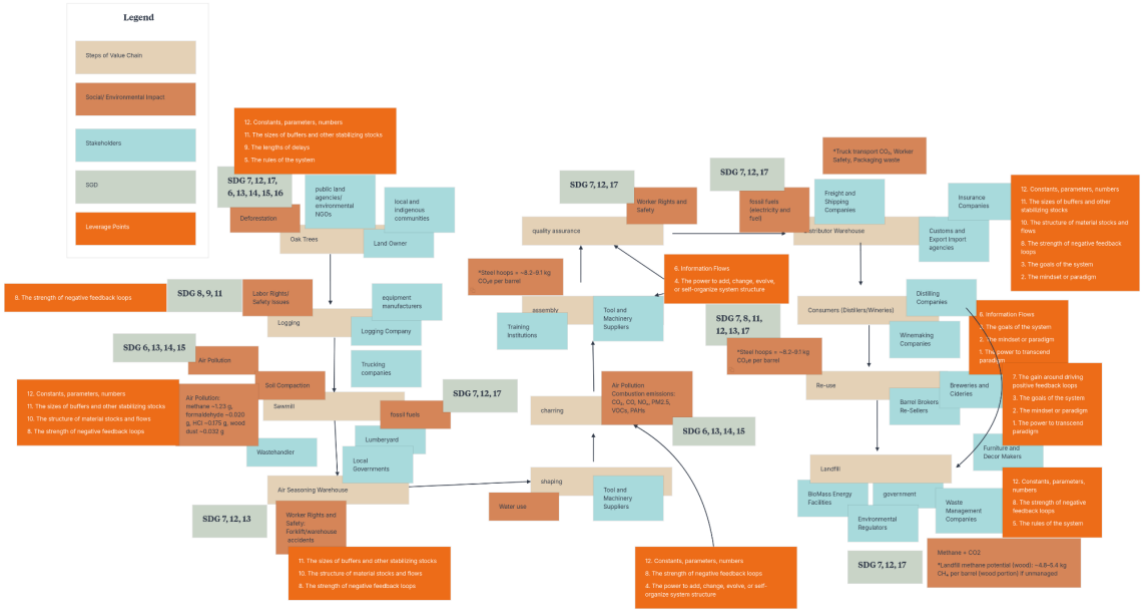
Soil Erosion

- Forest harvesting can also affect watershed dynamics. Road construction and improper harvesting may alter runoff patterns, which could increase the rate of sediment that ends in nearby streams. Sediment yield from inadequately managed logging roads can increase by two to ten times baseline levels. (USDA Forest Service, 2014).

3. SDG Impact Analysis

SDG	IMPACT	ANALYSIS
6 - Clean Water 7 - Affordable Clean Energy 12 - Responsible Consumption and Production	Safe and accessible resources, reduced carbon emissions,	• Implementation of practices and interventions that reduce water use and ensure pollutants are not released back into waterways • Considerations around alternative methods of charring that could reduce emissions
8 - Decent Work & Economic Growth 9 - Industry, Innovation & Infra structure 11 - Sustainable Cities & Communities	Ethical economic practices and community engagement	• Investment in the community beyond activities directly associated with production • Providing living wages to employees
12 - Responsible Consumption and Production 13 - Climate Action 14 - Live Below Water 15 - Life on Land	Rights of nature, waste generation, land pollution	• Increased efforts around deforestation prevention beyond negating our negative impact • Encourage biodiversity preservation efforts
13 - Climate Action 16 - Peace Justice and Strong Institutions 17 - Partnerships for the Goals	Collective action / community-based decision making	• Collaborate with others to maximize positive impact and reach • Research leaders in the areas of interest and implement proven practices

4. System Redesign Value Map



Leverage Point	Effected Block	Factors
12. Constants, parameters, numbers	Oak Tree, Sawmill, Charring, Distribution, Landfill	Harvest limits & replanting ratios. Air pollution limits. Combustion emission thresholds. Transport emission standards. Landfill emission limits.
11. The sizes of buffers and other stabilizing stocks	Oak Trees, Sawmill, Air Seasoning, Distribution	Sizes of buffers.
10. The structure of material stocks and flows	Sawmill, Distribution	Transport network design
9. The lengths of delays	Oak Trees, Air Seasoning,	Tree growth timeline. Multi-year seasoning time.
8. The strength of negative feedback loops	Logging, Sawmill, Charring, Distribution, Landfill	Logging practice. Byproduct left. Air-quality enforcement.

		Transport emissions. Disposal emissions
7. The gain around driving positive feedback loops	Re-use	Reusing barrels.
6. The structure of information flows	Assembly, Quality Assurance, Consumer, Re-Use	Barrel tracking system (QR or RFID) across supply chain
5. The rules of the system	Oak trees, Air seasoning, Landfill	Deforestation laws. Air pollution limits. Waste management laws.
4. The power to add, change, evolve, or self-organize system structure	Air seasoning, shaping, charring, assembly, quality assurance	Internal Processes
3. The goals of the system	Consumer, Distributor	Harvest and deliver as much product as efficiently as possible. (distributor) Obtain as much product as cheaply as possible (consumer)
2. The mindset or paradigm	Consumer, Distributor	Natural resources are inputs for value creation, and responsibility ends at point of sale.
1. The power to transcend paradigm	Consumer, Re-use	Barrels as a circular service rather than a one time sale.

5. Comparison of Business Sustainability Strategy Narrative

Biomimicry: The Honey Badger

To become a sustainable player in barrel production business, ISC needs to be resilient, adaptive, and efficient. To do so, ISC should follow the example set by the honey badger. Honey badgers offer a powerful comparison for designing a sustainability strategy that balances durability with flexibility, innovation with practicality, and strength with environmental awareness.

Thick Skin

ISC can protect the barrels' most easily damaged areas by adding chime guards as a way to mimic the honey badger's thick, rubbery skin that absorbs damage. The chime guards protect the integrity of the barrel during handling, transport, and storage, and they are sustainably made from recycled plastic and aluminium (Uniguard Machine Guards, n.d.). High-impact hoop coatings made from GreenKote reduce damage to the barrels. GreenKote is eco-friendly since it is free of solvents, emits no volatile organic compounds, and reduces the need for frequent recoating and replacement (GreenKote Ltd., n.d.). GreenKote has high corrosion resistance and does not need to be removed from the steel at the end of the life of the barrels. Implementing these changes connects to leverage point #7 because it extends the useful life of the barrels, reducing oak demand, replacement-related emissions, and repair waste.

Resource Intelligence

Honey badgers thrive in harsh environments because they extract value from limited resources, supporting a circular sustainability strategy, also connecting to leverage point #7. Waste is repurposed, not discarded. At ISC, 100% of the oak that is purchased is used for barrel production or converted into secondary products, reducing landfill use and reducing heating energy costs (Independent Stave Company, n.d.). The concept of barrels as a service reduces resource waste and encourages ISC to make barrels more durable. Durability also plays a role in resource intelligence since using long-lasting, locally sourced construction materials reduces transportation emissions and minimizes maintenance demands over decades. True sustainability is not only about reducing consumption today but preventing avoidable consumption tomorrow.

Climate Adaptation

The honey badger survives diverse climates by using burrows to regulate its temperature naturally (Skinner & Chimimba, 2005). This behavior demonstrates environmental intelligence rather than reliance on constant energy input. Applying this logic, the facility should incorporate passive thermal strategies. Thick, insulated walls and climate-responsive design create stable maturation environments with reduced mechanical heating and cooling needs. By designing with natural temperature regulation in mind, the business lowers long-term energy consumption, operational costs, and carbon emissions. This concept is tied to leverage point #4 because sustainability becomes embedded in the architecture itself.

Conclusion

The honey badger biomimicry model demonstrates that sustainability is resilient, adaptive, and strategic. By adapting ISC's business model to reflect the characteristics of the honey badger, it actively prepares for volatility, maximizes resource value, and builds infrastructure that can withstand environmental and market pressures. These sustainability strategies make IMC's sustainability practice a competitive advantage rather than a compliance measure. By adopting these biomimicry measures, ISC activates the power to transcend the paradigm of barrel production and sales by maximizing barrel life through a circular service rather than a one time sale.

6. Business and Product/Service Innovation Narrative

Our most impactful business strategy is switching from selling spirit barrels as a product to selling them as a service. This would mean instead of selling a set amount of barrels to a company, we would retain ownership and rotate them through different companies as needed. Although we do not have an exact number of barrels repurchased, Kentucky Cooperage LP, a subsidiary of ISC, has invested 4.1 million dollars into their used barrel operations (Distillery Trail 2022). Rather than treating barrels as a one-time transaction, International Stave Company would manage the entire lifecycle of each barrel. This shift changes our role from a manufacturer of wooden containers to a long-term asset manager responsible for performance, maintenance, and end-of-life recovery (Tukker 2004).

Under this model, our new oak barrels that have never been used would begin their journey at bourbon distilleries . After the aging process at the bourbon distillery, the barrels would be returned to International Stave Company for maintenance or transported directly to another cidery, distillery, or winery so they can use the bourbon barrels to age their product, as many already do. Instead of these secondary transfers happening informally in resale markets, we would coordinate and manage the rotation process. This would allow us to maximize the number of productive life cycles each barrel completes before reaching its true end-of-life.

Because we would retain ownership of the barrels, we would be incentivized to design them for longevity. Our goal would be to make barrels that last as long as possible across multiple aging cycles. This aligns directly with our sustainability vision. Oak trees suitable for cooperage take decades to mature, and the production of new barrels requires significant natural resources. By designing for longevity, we reduce pressure on forests and decrease the need for continuous raw material extraction. Barrels could be constructed with a stronger emphasis on durability, modularity, and repairability. For example, staves or steel hoops that become damaged could be replaced individually rather than discarding the entire barrel. Components would be assembled in ways that allow for easier disassembly and refurbishment, which in turn extends the usable life of each barrel.

Under the new system, barrels would cycle back to ISC after each use to be inspected, serviced, and repaired if needed before being redeployed to the next distillery. Broken staves or steel rings

could be fixed using parts from other broken barrels to reduce waste. This approach not only reduces waste but also lowers production emissions associated with manufacturing new barrels. Once barrels reach their true EOL and can no longer be safely used for aging spirits, any parts that can be reused would be set aside. The remaining wood could be repurposed into furniture, decorative pieces, smoking chips, or other products, and the steel hoops would be recycled. This ensures that materials remain in circulation for as long as possible.

In addition to environmental benefits, this model significantly changes our revenue structure. Instead of generating revenue through a single transaction, we would be creating a revenue stream through leasing agreements or a subscription service. Customers would pay for access to high-quality barrels and the service of maintaining, replacing, and rotating them. This provides more predictable and stable cash flow for International Stave Company and reduces reliance on fluctuating lumber prices. By retaining ownership, we can capture value from turning an informal barrel resale market into a formal part of ISC business and environment strategy, which gives more complete control of all the barrels.

There are also important strategic risk considerations, with one major risk being customer resistance to changing from ownership to a service model. Many distilleries are accustomed to purchasing ownership of the barrels outright, and shifting to leasing agreements may require education, contract restructuring, and trust-building. Another risk involves organizing the collection, inspection, repair, and redistribution of barrels from many distilleries. To mitigate this, barrels would be digitally tracked, allowing ISC greater visibility and control over the barrels' inventory, condition, and location. This tracking system would also allow us to monitor usage cycles and forecast refurbishment needs. There are also supply chain risks associated with this system. Climate change, pests, and land-use pressures threaten white oak forests (Espinoza, 2020). By extending the life of each barrel, we reduce exposure to raw material scarcity and price volatility.

By redesigning how we deliver value, rather than simply improving how we manufacture barrels, we align our business incentives with our sustainability goals. The longer a barrel lasts, the better it is for the environment and for our bottom line. This alignment makes the "barrels as a service"

model our most impactful strategy for achieving our vision of a more sustainable and resilient International Stave Company.

7. Types of Goods and Market Failures Narrative

Position in the Goods Matrix

ISC operates in a variety of classes of goods, which illustrates how a company can engage itself in redefining the resource classification using property rights (Ostrom, 1990). Oak timber is the main input at ISC that is a natural resource (rival and non-excludable). Kiln-drying and steaming water is one of the most common resources and one that cannot be excluded completely, and ISC needs to approach regulators and nearby users carefully. Metal hardware part is a market commodity, which can be competed with and excluded by conventional market dealings.

In the output side, the charred white oak barrels of ISC are market goods, which are rival and excludable, and their prices and distribution are based on transactional relationships. More strategically noteworthy is the fact that ISC is turning manufacturing waste (sawdust, wood chips, off-cuts), into marketable mulch and biomass fuel (Independent Stave Company, 2022). ISC establishes market values on what would otherwise go to waste by introducing excludibility and capturing value on the materials that would otherwise be a non-rival social bad in the form of waste disposal pollution. This change is based on one of the fundamental concepts of a circular economy, the redesign of waste as a product that is being sold instead of a cost that is being managed (Geissdoerfer et al., 2017).

Externalities: Favors and Disfavors.

ISC creates a set of externalities that costs and benefits are on ones who are not directly engaged in its market dealings. On the negative side, the local pollution and greenhouse gas accumulation through air emissions in the charring of barrels and loading into kilns has to be paid by the local communities and the global climate as opposed to the price of barrels. The additional unpriced social costs are the risk of water discharge and the possible health exposures of workers to sawdust and chemicals and the disruption of the ecosystem by harvesting activity. Of particular concern to worker safety is OSHA referencing ISCs Lebanon, Missouri facility when it was noted that five injuries resulting in amputation of employees took place within 14 months, which shed light on machine guarding and in hazard control (U.S. Department of Labor, 2019). One downstream externality should not be overlooked as well: since ISC is the main supplier of the bourbon industry, it indirectly facilitates alcohol drinking and all social expenses related to it, including health consequences and addiction (Centers for Disease Control and Prevention, 2024;

World Health Organization, 2024) although multiple market processes may lie between barrel production and the final customer. Simultaneously, spirits have significant positive externalities, serving as a cooking ingredient, as the basis of traditions in cultures, and as an engine of regional tourism economies (Kentucky Distillers' Association, 2024).

ISC generates positive externalities, which are significant and meaningful against these. The benefits of its reforestation initiatives, planting trees on multi-decade cycles create the carbon sequestration, biodiversity, and watershed protection benefits that stretch way past its property (Independent Stave Company, 2022). The ISC offers permanent manufacturing jobs, supplier contracts, and tax revenues to the local communities (Kentucky and Missouri) in rural areas where facilities are located, which drive the overall economic growth.

Positioning of Public-Private Value Matrix.

ISC is located well in the high market value position, it is a profitable, dominating force in a niche international market. Its public value positioning, in the real sense, is actually controversial. The argument of positive public value is based on environmental care by ISC (Independent Stave Company, 2022), economic impact on rural areas, and maintenance of the old traditional industries having cultural values. The bourbon business is a symbol of the American cultural identity and makes a major contribution to the tourism sector, where Kentucky bourbon tourism is bringing a considerable economic effect on the area (Kentucky Distillers' Association, 2024). The reason why the social value is lower mainly rests on the fact that the ultimate use of the products produced in ISC is barrels that are used to produce bourbon, social costs of alcohol consumption, addiction, impaired driving and are actually a real loss of societal welfare that is facilitated by the lowering of the alcohol supply chain (Centers for Disease Control and Prevention, 2024; Sacks et al., 2015). ISC is on the high market value / moderate-to-low public value quadrant, which is the most defensible. The company is not just low public value, its forestry business and economic activities are real. Even though alcohol has a strong cultural and heritage-related meaning, especially the strong ties between bourbon and the tradition of the American people, the adverse health and social implications of its use do not allow ISC to boast of a high level of public value.

8. What is a business for

Core Purpose: Profitable Barrel Manufacturing

Independent Stave Company's primary function is to provide high-quality oak barrels and staves for the global wine and spirits industry. On a superficial level, Independent Stave Company is a for-profit business that generates revenue through sales, reinvests in mills and cooperages, and competes based on quality and customer relationship-building. In this sense, ISC's function is to provide for-profit business operations: the first bottom line.

Beyond Shareholder Primacy

However, businesses do not exist solely for quarterly profitability. In fact, the idea that corporations must maximize shareholder value is a management philosophy rather than a legal requirement. As a family-owned business, ISC has more flexibility in terms of operating horizons than publicly traded corporations that must respond to quarterly earnings expectations. Since its primary input is white oak trees that regenerate over a period of decades rather than years or quarters, the business cannot operate in a way that maximizes profitability without threatening ecological viability. In fact, the business's very existence depends on keeping forest ecosystems healthy so that they can provide inputs for barrels. Thus, ISC's function extends beyond for-profit business operations to also include long-term stewardship of a key natural resource.

Managing for the Triple Bottom Line

From a triple bottom line perspective, ISC must balance profit, people, and planet. Profit is required for business operations and investment in mills and cooperages. The "people" component includes occupational safety and health in forestry and cooperage work, community stability in rural areas where cooperages are often located, and relationship-building with natural resource suppliers. The "planet" component is inherently connected to the business model of ISC, in that forest regeneration, biodiversity, and climate change are not just desirable outcomes but rather are critical to the operation of business in general. Thus, sustainability is not altruistic in nature, but rather it is operating in one’s own self-interest and living in a state of ecological interdependence.

Positioning on Defourny and Nyssens' Typology

Consistent with Defourny and Nyssens' typology, ISC is primarily embedded in the capital interest domain in that it is a market income, privately owned enterprise that earns its revenues from competitive sales. However, shareholder primacy is not legally required and represents a managerial choice rather than an institutional necessity (Stout, 2012). The increasing salience of forest stewardship, certification, and sustainability governance suggests the possibility of a new form of organizing that combines economic interests with public interest goals (Battilana & Lee 2014). The legitimacy and sustainability of the ISC will not only depend on customers but also on regulators, forest landowners, rural people, and environmental groups. In this sense, the ISC is at the crossroads of economic value creation and public ecological responsibility. Being an economic organization that seeks profit maximization, the sustainability of the organization is closely related to the larger systems it operates in.

Structural Barriers

However, there are barriers that the ISC needs to consider for its sustainability initiatives. The company is faced with cost pressures, infrastructure lock-in effects, supply chain fragmentation, and weak market signals to internalize environmental externalities. In the absence of reporting guidelines or quantified climate objectives, some of its material impacts are undermeasured. However, these constraints do not diminish ISC's purpose; they clarify the managerial decisions that can help the company overcome shareholder primacy.

Conclusion: What ISC is ultimately for

Independent Stave Company's ultimate purpose is to do more than merely produce barrels. It is to sustain the economic, ecological, and social systems that enable barrel production. ISC's long-run purpose is to generate sustainable value not just to its owners and customers but to the forests, workers, and communities on which its very existence depends.

9. Assessment and Reporting

ISC does not publish a standalone, comprehensive sustainability/ESG report on its website like many companies do. Instead, ISC's sustainability communication is largely website-based through public press releases and focuses on forestry-related certification, partnerships supporting long-term oak supply, and qualitative descriptions of operational practices, like using "100% of the oak" they purchase by turning residual wood into internal fuel or other by-products (Independent Stave Company, n.d.-c). This approach fits reporting that can function as "certification-driven reporting" rather than a full performance reporting.

ISC's third party certification and documented certificates are generally credible because they are externally defined standards with audit and verification systems in place and commonly used in the industry like the Programme for the Endorsement of Forest Certification (PEFC), Sustainability Forestry Initiative (SFI) Fiber Sourcing Standard, and ISO 14001 environmental management certification (ISC, n.d.-c; ISC, 2022). The credibility problem is not that the certifications are fake but that the overall picture is not transparent, complete, comparable, or verifiable as a sustainability report. Under GRI reporting guidance, high quality sustainability reports should follow principles such as accuracy, balance, comparability, completeness, and verifiability, and should clearly identify and disclose material topics and management approaches. Since ISC's sustainability information is largely selective and unqualified, it does not provide full accounting of key impacts and risks across the value chain. This is a warning that reporting can increase greenwashing risk if outcomes are not shown.

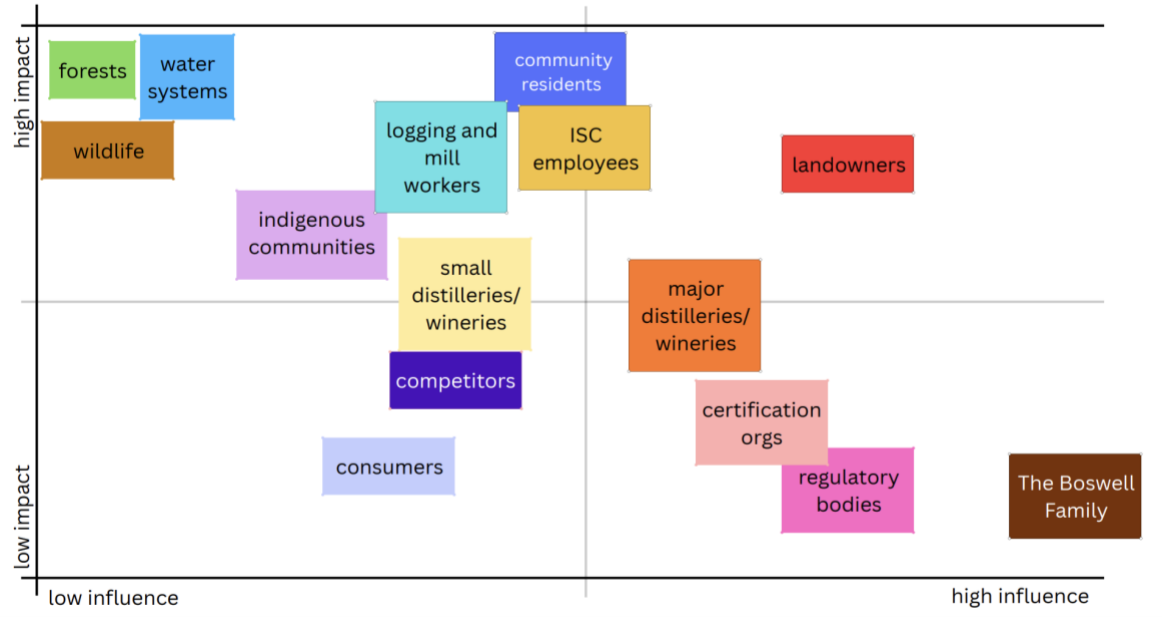
ISC should add true sustainability reporting so its claims become standard and comparable. They should use GRI Standards to report impacts on people and planet through materiality processes grounded in stakeholder engagement or use SASB to report investor/financial sustainability metrics in a consistent way by the industry. Seguin Moreau (n.d.) uses carbon auditing which quantifies CO2 footprints and ISC should consider implementing such audits to be more quantifiable.

ISC's sustainability endorsements come from forestry certifications and general ecological commitments but does not fully disclose several impacts that appear across the barrel value chain. The largest gap is the absence of clear, quantified information about operational resource

use. For example, the company does not publicly report detailed data on water consumption, fossil fuel or energy use, or emissions throughout its operations. These disclosures would be organized by Scope 1, 2, and 3 so that the stakeholders have better insight into significant environmental impacts that occur in the value chain. Non-wood materials used, such as steel hoops and other purchased materials have their own sustainability effects on emission and labor conditions and are barely mentioned in ISC's sustainability communications. ISC emphasizes certified or sustainably managed forests, but these frameworks don't fully address long-term issues such as oak regeneration, forest resilience, wildlife habitat, and the ecological systems that depend on oak forests. On the social side, there is very little disclosure on labor safety data across their whole supply chain (logging, milling, cooperage work, and distribution). These data on injury rates, safety training, etc. would actually help stakeholders better understand the working conditions in the supply chain. ISC also rarely discusses the downstream impacts of the liquor industry that its barrels support. Distilling itself is resource intensive (energy, water, waste) and alcohol consumption carries broader public health effects. These issues create reputational risks even if they occur outside the company's direct control. However, recently, ISC has created a dedicated Sustainability Division to set measurable goals and improve systems indicating efforts for stronger future reporting standards (ISC, 2025).

Overall, evidence suggests that ISC faces some risk of greenwashing through vague/incomplete disclosure, albeit, there is not enough evidence to claim intentional deception. The concerns come from the gap between positive claims, like reducing carbon impact or sourcing wood responsibly, and the limited availability of comparable performance data across most material environmental and social areas. Global Reporting Initiative (2021) says that reporting standards emphasize credibility and verifiability, so the absence of detailed metrics can weaken stakeholders' trust. Albeit, ISC's wood sourcing is partly supported by industry recognized certification systems (PEFC and SFI certifications). These certifications make any fabrications unlikely but do not replace full transparency. This could be interpreted as greenhushing as ISC engages in more selective disclosure. To reduce these risks, ISC could move beyond general claims and certifications by publishing a detailed sustainability report that addresses operational boundaries, Scope 1, 2, and 3 estimates, disclose safety indicators, and report biodiversity management practices related to forests that the company depends on. This will help recognize ISC's efforts and actions that support environmental improvements rather than industry trends.

10. Stakeholders and Rights Holders Narrative



Independent Stave Company stakeholders fall into four primary categories: rights holders, operational partners, customers, and financial and strategic actors.

<u>Rights Holders</u>

Brodeur et al define rights holders as individuals whose human rights can be directly impacted by activity of the company (Brodeur et al., 2023). For our company, this includes white oak forests, wildlife, and surrounding water systems which are materially affected by harvesting, milling, and transportation. These natural systems experience high impact but ultimately have no decision-making power. Using the stakeholder influence–impact matrix described by Wiek et al. (2011), they fall into the high-impact/low-influence quadrant. Local communities in logging regions across Kentucky and Missouri also qualify as rights holders. Mills and harvesting operations provide employment, but they can also impact landscapes and natural systems. Another essential rights holder group is Indigenous communities whose ancestral lands overlap with these forests. Tribes connected to Kentucky and Missouri, including the Osage Nation and the Shawnee and Cherokee tribes, maintain longstanding cultural and spiritual ties to these lands even though they may not have legal title (Oklahoma Historical Society, n.d.; West Virginia University Libraries, n.d.). If ISC is pursuing holistic stakeholder engagement, they must address historic power imbalances between the business and tribes with ties to these lands.

Operational Partners

Operating partners include private forest landowners, logging contractors, truckers, sawmills, steel suppliers, and employees across ISC facilities. Landowners are high impact, but also high-influence as they have agency over entering contracts with ISC. Employees fall in the mid-high impact as they are directly impacted via their work environment. They fall into the middle to low influence category because as individuals, they have limited influence over the system, although unionization would increase their leverage. Regulatory bodies such as the Alcohol and Tobacco Tax and Trade Bureau (TTB) carry medium influence as they define production standards that the company must adhere to. Certification organizations like the American Tree Farm System fall into a similar category as they influence forestry practices. Both regulatory bodies and certification organizations experience low impact from the activities of the company.

Customers

ISC's customers include several major bourbon producers, wineries, cideries, and other cooperages (Distillery Trail, 2022). Some of their highest-influence American clients include Heaven Hill (Elijah Craig) and Brown Formal (Jack Daniels) who wield substantial power and sway in the industry. In the impact matrix, we have placed customers as relatively low because they experience very little direct impacts from the operations of the company. On the other hand, because ISC holds such a large market share of white oak barrel production, the decisions of the company do have an impact on these brands. On the influence matrix, the customers exert a low-moderate influence over the company. Consumers of these brands increasingly expect companies to meet higher sustainability and ESG standards, and these expectations can place pressure on suppliers like ISC to adjust their production practices (PwC, 2021). As bourbon and wine production expands globally, it also puts higher demands on ISC as well.

Financial and Strategic Actors

Since its founding as the TW Boswell Stave Company in 1912, ISC has continued to be owned and operated by members of the Boswell family (Independent Stave Company, n.d.). Members of the family (shareholders) fall into the high influence/low impact quadrant of the matrix as they don't experience exposure to the day to day operations of the business, but hold the most direct decision making power over the actions of the business. Competitors, including other cooperages and vertically integrated distilleries, like the Brown-Forman Cooperage and The County

Cooperage are important stakeholders. Their business operations indirectly influence ISC through market pressures and competition within the industry. These stakeholders fall into a moderate/low impact and influence. Finally, advocacy organizations, climate activists, and anti-alcohol groups represent moderate to low impact and influence stakeholders for ISC because although they may not engage directly with ISC, they can influence its social license to operate.

Recommendations

Stakeholder engagement centers on relationship building and navigating power differentials (Wiek et al., 2011). ISC does engage in a meaningful way with landowners in Kentucky and Missouri through its Log Procurement and Land Assistance Programs (Independent Stave Company, n.d.-a; Independent Stave Company, n.d.-b). Participants of these programs receive support from ISC through land management plans, replanting, and sustainability certifications. However, their engagement with the communities they mill and manufacture in is basically a one way street. While initiatives such as the Volunteer Firefighter Initiative and research donations build goodwill, they are sad substitutes for real channels for community input in decision-making (Independent Stave Company, 2020; Independent Stave Company, 2021). Meaningful participation is predicated on early consultation, accessible information, and real opportunities to influence outcomes (Brodeur et al., 2023). Without these essential processes, high impact stakeholders remain low influence.

ISC must expand who is formally included in its engagement processes. It would benefit ISC to create a regional multi-stakeholder forestry council with tribal representatives, community leaders, and ecological experts to reduce power asymmetries and strengthen feedback pathways. In addition, ISC should collect operation and performance metrics such as forest regeneration rates, community satisfaction, and employee wellbeing, and make them public because as Brodeur et al. note, “Access to information is a prerequisite for participation” (2023, p. 5). Together, these steps would rebalance ISC’s stakeholder ecosystem and strengthen its network of relationships.

11. Organizational Change

Although sustainability is a stated goal, several forces are preventing the company from becoming more sustainable. Of these forces, those categorized as coercive pressures are arguably the most prominent. The bourbon barrel industry is highly regulated, which limits the company's ability to experiment with alternative materials that might be more sustainable. Beyond that, their sustainability efforts are limited by adherence to the standards set by the environmental organizations that provide our compliance certificates. While these certificates demonstrate compliance and reliability, they limit ambition and cause ISC to perform to stated metrics instead of their own aspirations. In ISC's case, these coercive pressures come from governing forces and seem to be followed as a matter of requirement instead of choice.

Mimetic pressure also plays a role. In the bourbon barrel industry, ISC is in the unique position of being an industry leader. As a leader, they are given the flexibility to make decisions that set the standard for others to follow. In the industry, no major bourbon barrel company, including ISC, publishes detailed sustainability and emissions reports. In the same vein, no one is disputing how barrels are made or challenging the rules that have been in place since 1938. When no one in the industry is pushing the boundaries, it is easy to stay complacent. If others are not making reports, and reports might disclose more negatives than positives in practice, it is easy to fall into the mindset of "why should we?"

The most subtle of the pressures are those within the normative category. The barrel industry is built on tradition and buttressed by the art of craftsmanship. While these practices lend themselves to a certain caliber of quality, they also create a sense of resistance to change. As a family-owned company, ISC likely has been refining its craft and has passed down tips, tricks, and practices from generation to generation. It is unlikely then, that a younger generation might question or push back against long-standing processes. Because these norms are so deeply embedded, challenging them might feel like challenging the family company's legacy.

Although these pressures exist, overcoming them is possible. First, as leaders, ISC can redefine what success looks like. By expanding metrics and looking beyond strictly KPIs, they can shift things like emissions and environmental impact into their business priorities. Additionally, if

they were to adopt standard reporting regulations and publish reports regularly, they would be letting the public hold them accountable. Leadership could also push to raise the standard at an industry wide level, rather than simply meeting what has traditionally been accepted. By focusing on the interests of shareholders, instead of strictly stakeholders, ISC could broaden their understanding of what responsibility and authority look like for others impacted by their practices.

Entry level employees and younger generation family members can make a difference. By holding the company accountable to ensure they are actually doing what they say they are doing, they can make sure ISC is not undermining its own commitments. Entry level employees can advocate for better sustainability practices by respectfully raising concerns, suggesting improvements, and looking globally to see how others might be setting an example. If new employees can work with leadership and encourage them to expand how they define success, ISC could move from complying with standards to setting them.

12. Recommendations

1. **Barrels as a service:**

   ISC could tap into an underutilized segment of the market by turning the finished product (the barrel) into a service that can be bought and sold. In this scenario, the barrel would be sold in a "rental" capacity, where buyers will lease the product for a predetermined period of time with the expectation it will be returned to ISC at the end of the rental period. Upon its return, regular maintenance will be performed to prolong the life of the barrel and will be leased again by the same or another buyer.

2. **Alternative for steel hoops:**

   ISC should commit to using exclusively recycled steel products for barrel hoops. Of the materials used in ISC's barrels, the hoops account for the largest rates of emissions, as steel production is one of the most carbon intensive processes, due largely to the rate of coal combustion during steelmaking. By utilizing exclusively recycled steel, ISC could reduce steel related emissions by up to 75%. (World Steel Association, 2022; IEA, 2023).

3. **Greenkote steel sealing:**

   We encourage ISC to use high-impact hoop coatings made from Greenkote to reduce damage to the barrels. Greenkote is eco-friendly since it is free of solvents, emits no volatile organic compounds, and reduces the need for frequent recoating and replacement. Implementing this change extends the useful life of the barrels, reducing oak demand, replacement-related emissions, and repair waste. Greenkote has high corrosion resistance and does not need to be removed from the steel at the end of life of the barrels (GreenKote Ltd., n.d.).

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Appendix A

PESTLE Analysis

This appendix provides the external macro-environmental analysis (PESTLE) conducted for Independent Stave Company (ISC) as part of the risk and strategic assessment. The PESTLE framework evaluates Political, Economic, Social, Technological, Legal, and Environmental factors that influence ISC's barrel manufacturing business and long-term sustainability. This analysis informs the Vision & Risk narrative and subsequent strategic recommendations in the main report.

P	Political
• Forestry regulations and land use policy – Government restrictions on hardwood harvesting in the U.S. and France could constrain raw material supply. • U.S.–E.U. trade tensions can reintroduce tariffs on oak barrels, directly affecting demand and margins.	

E	Economic
• Because sales are worldwide but mills are only located in the U.S. and France, currency differences could pose a challenge/risk due to the high volatility of currency exchange, causing many other markets to turn to more local products. • Oak wood pricing and shortage risks influence production costs; limited availability due to forest regulations or supply chain issues can increase costs. • Economic growth or recession in key markets (such as the U.S., Europe, and Asia) affects demand for premium alcoholic beverages. • Beverage industries often have long production cycles, leading to cyclical demand for barrels.	

S	Social

- Decline of alcohol consumption worldwide.
- Lack of skilled workers: limited pipeline of trained craftspeople.
- Family-owned structure carries risk of culture and production changes if the company were to be sold.

T	Technological

- Automation in the lumber supply chain.
- Substitution technologies – alternative aging methods (oak chips, staves, rapid-aging tech) threaten traditional barrel demand, especially in cost-sensitive segments.
- Lab analytics and chemical profiling are becoming commercial tools (not just R&D); GCMS analysis on oak samples and "organoleptic profile matching" are used to mirror a desired sensory profile.
- ISC partnered with the University of Kentucky and Maker's Mark to map the white oak genome, funding research to support long-term supply.
- Supply chain traceability technology is becoming mandatory; regulations like EUDR (see Legal) are forcing digital due-diligence statements and geolocation-driven traceability systems.
- Process-control technology is shifting cooperage from "craft-only" to craft-and-data, with repeatability emerging as a key differentiator.
- Distillers increasingly demand repeatability (consistent flavor outcomes) and documentation, especially as premiumization and brand consistency become critical.

L	**Legal**

- ISC operates within a complex, multi-layered legal environment spanning labor and employment law, occupational safety, environmental and forestry regulation, product liability, trade law, and waste management.
- Legal risk is concentrated upstream in logging and milling (worker classification, safety compliance, environmental impacts) and downstream in product liability, sourcing transparency, and ESG-related disclosure obligations.
- Forestry certifications (FSC, SFI, PEFC) function as quasi-legal requirements once embedded in contracts with customers and suppliers – loss of certification can trigger contract termination, reputational damage, and loss of market access.
- Forest Stewardship Council (FSC) prohibits illegal logging and requires protection of biodiversity, respect for worker and community rights, and long-term forest viability; enforced through third-party audits.
- Sustainable Forestry Initiative (SFI) emphasises conservation, biodiversity, water quality, and responsible harvesting; subject to criticism for allowing intensive practices such as clear-cutting under certain conditions, elevating ESG and reputational risk.
- Programme for the Endorsement of Forest Certification (PEFC) verifies sustainable forest management through independent audits and chain-of-custody traceability, supporting compliance with illegal-logging and sourcing transparency laws.
- OSHA (29 CFR §1910.266) governs workplace safety for logging and sawmill operations, requiring safety training, qualified operators, safe work practices, PPE, and hazard avoidance.
- Tree felling rules (§1910.266(h)) prescribe proper notch and back-cut techniques, pre-cleared escape paths, minimum worker spacing, and prohibition on working under lodged trees.

- Structural safety risks include logging contractors lacking mill-employee protections, lower inspection frequency in rural operations, and chronic hazards (wood dust, noise) causing long-term harm.
- Logging sector unionization is extremely low due to independent contractor status and fragmented workforce; reduced collective bargaining increases reliance on regulatory enforcement rather than negotiated protections.
- Sawmill sector unionization is slightly higher (~7.8% nationally) but still limited; labor disputes are localised but can disrupt production and raise compliance scrutiny.
- Under U.S. federal law (TTB), bourbon must be aged in new, charred oak containers made from oak wood only and free from additives – barrel non-compliance can render a customer's product legally non-bourbon, exposing both parties to liability.
- Wood species must be oak (typically American white oak); interior must be charred (not toasted only); and bourbon law requires new barrels for each fill – reused barrels marketed for bourbon aging violate federal standards.
- Barrels function as food-contact materials; prohibited chemical treatments or contaminants that could migrate into the spirit can trigger product recalls, civil liability, and reputational damage.
- Distillers rely on cooperages to certify barrel compliance (wood origin, new-barrel status, charring process); inaccurate documentation can lead to enforcement actions, breach-of-warranty claims, and loss of customer trust.
- Barrel specifications are typically embedded in supply contracts; failures may result in barrel rejection, lost aging time, and indemnification claims – long aging periods magnify downstream legal risk.

E	Environmental

- Raw material scarcity through deforestation and forest cover loss.
- Biodiversity loss and ecosystem degradation – degraded forests are less resilient and produce lower-quality timber.
- Climate change affects forest health and drying times, and also affects supply stocks.
- Wildfires, pests, and disease outbreaks pose risks to timber supply.
- Water scarcity impacts forest management and mill operations.
- Land-use competition and agricultural expansion reduce available forestry land.
- Market premium for sustainable sourcing creates a commercial incentive for responsible forestry practices.
- Growing demand for resource-efficient and zero-waste products aligns with ISC's operational approach.
- ISC's French oak stave mill, Merrain International, is certified through PEFC, as are the French oak barrels crafted in the U.S. and France, verifying that French oak is sourced from sustainably managed forests.
- Domestic stave mills and cooperages hold Sustainable Forestry Initiative® (SFI) certification; all American oak barrels manufactured domestically are certified to the SFI Fiber Sourcing Standard.
- ISC utilises 100% of the oak it purchases – wood not used in staves, heading, or oak alternative material is used internally as fuel or sold as high-quality by-products for use in other industries.